

DCS-CTN: Subtle Gesture Recognition Based on TD-CNN-Transformer via Millimeter-Wave Radar

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Abstract—Gesture recognition has been a hot research topic in human–computer interaction, since contactless gesture recognition will provide increasing applications in many fields. Millimeter-wave (mmWave) radar well serves this technology because of its high accuracy, easy integration, and strong anti-jamming ability in moving object detection. However, it is still challenging to meet the requirement of high precision in subtle gesture recognition based on traditional methods via point cloud or Range-Doppler heat map of mmWave radar. Considering the raw data from mmWave radar with more information, such as phase, we propose a system that uses the constructed mmWave radar data cube sequence and timedistributed-CNN-transformer network (CTN), called DCS-CTN system, to get higher hand gesture recognition accuracy. In this system, we introduce a time-distributed wrapper (TD) and convolutional neural network (CNN) to extract local features of the data cube sequence, a position encoder to retain time information of the sequence, and a transformer network to get global features of the sequence. The experiments results show that this system can achieve hand gesture recognition accuracy of 99.75%, which is significantly higher than the other traditional approaches.

Index Terms—Gesture recognition, human–computer interaction, millimeter-wave (mmWave) radar, transformer network.

I. INTRODUCTION

HUMAN computer interaction is critical to scientific and technological life for human, among which gesture interaction has broad applications, such as smart homes, somatosensory games, wearable devices, etc. Since gesture is one of the most expressive natural interaction behaviors of human beings, the design of interaction system based on gesture recognition is an important research direction in the field of human–computer interaction [1]. At present, most of the mature gesture recognition technologies are based on vision and inertial sensors. However, vision-based gesture recognition schemes are greatly affected by illumination conditions and may also leak personal information about users [2], [3], [4]. The recognition scheme based on inertial sensors requires users to wear additional sensors on hand, which is inconvenient and seriously affects the users’ experience. Since

millimeter-wave (mmWave) radar has the advantages of high spatial resolution, all-weather, easy integration, and privacy protection, the radar-based gesture recognition systems do not need additional sensors and are not affected by illumination conditions. Compared with the gesture recognition systems using low-band wireless signals, the mmWave radar sensor-based systems are easy to miniaturize and can be embedded inside a small device to greatly improve integrability and reliability of the device [5]. At the same time, mmWave signal has a stronger ability to capture small movements due to its extremely short wavelength, which can effectively identify subtle movements of fingers to achieve more refined interactive operations widely used in wearable devices, automotive assistance systems, and other applications that require higher interaction accuracy. Therefore, gesture recognition via mmWave radar is one of the most promising ways for mobile human–computer interaction.

Driven by the aforementioned motivations, there are a lot of works dedicated to gesture recognition using mmWave radar [6], [7]. Although traditional gesture recognition based on Range-Doppler heat map or point cloud has achieved satisfactory results in coarse-grained action predictions, it does not perform well in subtle gesture recognition. For gesture recognition based on point cloud, when the received radar signal is processed into point cloud, some useful information will be lost at the same time [8]. For gesture recognition based on Range-Doppler heat map, some advantages of multiple antennas of MIMO radar are not fully taken, and angle information of target is missing. To deal with these problems, we use the data cube sequence of radar for gesture recognition in this article. Precisely, we capture the low-voltage differential signaling (LVDS) of millimeter-wave radar echo to be processed into radar data cube. Since a gesture is a continuous action, we arrange each frame of radar data cube of a single gesture into a sequence in time order. Then, position encoding is used in our gesture recognition to retain its time information. We propose a scheme for the gesture recognition using the CNN-transformer network (CTN) algorithm with higher accuracy than the traditional approaches. At present, there are more and more scenarios in which people use graphics processing units (GPUs) for computing. The CTN used in this article can make full use of the parallel computing capability of GPU by virtue of a proposed special structure of transformer. The main contributions are summarized as follows.

- 1) We build up an mmWave radar testbed to capture three different data of gestures at the same time, i.e., LVDS of radar echo, point cloud, and Range-Doppler heat map.

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The testbed has two threads. One thread is used to collect the LVDS of radar echo, and the other one is used to collect the point cloud, and Range-Doppler heat map of the radar. Moreover, we build our data sets characterized by the LVDS of radar echo, the point cloud, and the Range-Doppler heat map. Each data set is constructed with more than 20 kinds of subtle gestures from 20 participants.

- 2) We propose a unique data processing method to process the radar echo into radar data cube sequence. Point clouds are obtained based on a constant false alarm rate (CFAR) method, which are normally very sparse. Since a Range-Doppler heat map only contains range and velocity information of a target, it may degrade the performance of gesture recognition. Considering these problems, we propose a data processing method to process the received radar echo into a data cube sequence which is dense data with both distance and velocity information to well use 3-D space information of gesture movement to realize gesture recognition with high accuracy.
- 3) In this subtle gesture recognition based on millimeter-wave radar, we propose an attention-based network CTN which combines TD-CNN with transformer to achieve a high gesture recognition accuracy. For the data cube sequence, TD-CNN extracts the features of the data cube at each time step, and the extracted features are still sequences with time information. We further encode the features based on a position embedding, which is used in the transformer layer to leverage their sequential order. In this way, the time information of each data cube is fully utilized. Experimental results show that CTN has a recognition accuracy of about 99.75%, higher than convolutional neural network (CNN) and CNN-LSTM when using data cube sequences as input features.

The remainder of this article is as follows. In Section II, we analyze some important works related to mmWave radar gesture recognition. In Section III, we present our main contributions of this study for gesture recognition. In Section IV, we provide the experiment results. Finally, we conclude our work.

II. RELATED WORKS

Gesture recognition has been investigated in many directions under different platforms, such as Wi-Fi sensing, sonic and ultrasonic sensing, vision-based sensing, infrared 3-D structured light sensing, and wearable sensing.

Wi-Fi Sensing: Before the appearance of mmWave radios, some researchers utilize Wi-Fi signal to realize various gesture recognition with commodity devices. There are many studies on monitoring human body postures [9], [10], [11]. Recently, to recognize hand gestures, the existing studies can be divided into two categories: 1) unique gesture recognition for different people [12] and 2) same set of gesture recognition [13]. However, most of them fail to detect the gestures with a “finger-grained” amplitude. They can only sense large and simple movements of a whole hand, such as pushing a

hand along a straight line. However, Wi-Fi sensing can perform well, but it is too sensitive to any surrounding change, which is a vital problem in real-world applications.

Sonic and Ultrasonic Sensing: Sonic and ultrasonic sensing consume low power because they have a low frequency. Most studies utilize the Doppler effect to sense gestures and to track hand movement. Still, these two kinds of waves are sensitive to ambient noise and their recognition precision is lower than using mmWave sensing.

Vision-Based Sensing: The principle of gesture recognition based on vision is usually to capture hand images using a camera, and then to extract gesture features for recognition by processing and analyzing these images [14]. It can recognize gestures with quite a high accuracy [15], [16]. However, it fails in darkness and non-line-of-sight scenarios and has a risk of privacy leakage. Generally speaking, vision-based gesture recognition inevitably collects image data from other parts of body when collecting gesture data, which poses a risk of exposing the user’s portrait privacy [17].

Infrared 3-D Structured Light Sensing: Infrared light sensing (shorter wavelength than mmWave’s) can finely model a hand movement with high precision to achieve gesture recognition. However, it has weaknesses of high energy consumption, vulnerability to blockages, and also possibility of privacy leakage.

Wearable Sensing: Compared to contactless sensors, wearable sensors can recognize more detailed gestures and capture more detailed information when used for gesture recognition to achieve higher accuracy. At present, there have been many significant studies in this field. Some researchers use wearable gloves equipped with RF-chipless tags with artificial neural networks (ANNs), or gloves with bimodal capacitive sensors for gesture recognition. Some researchers begin to use “data glove” for gesture recognition. All the above studies have achieved high accuracy in gesture recognition [18], [19], [20]. In addition, for gesture recognition based on wearable gloves, there is also a gesture recognition algorithm based on electromyography (EMG) and pressure signals, which achieves an accuracy of 96.05% for 21 gestures [21]. Although gesture recognition based on wearable gloves can obtain high accuracy, it has some limitations. Precisely, using wearable devices can cause inconvenience and reduce movability to users, and even make them feel tired and uncomfortable after a long time of use. On the other hand, the recognition range of gesture recognition is not large enough to recognize a wide range of gestures, which limits its applications for gesture recognition.

mmWave Sensing: In recent years, with the fast development of mmWave Radar techniques, gesture recognition based on mmWave Radar has become emerging, which can be roughly divided into three steps [22]: 1) collecting user’s dynamic gestures information based on millimeter-wave radar sensor; 2) preprocessing echo signals to extract dynamic gesture features; and 3) classifying gestures based on the extracted features. In hand gesture detection, mmWave Radar has the advantages of short antennas and wavelength, which makes a radar sensor smaller and easier to integrate into a smart device and performs better. NVIDIA develops a mono-pulse radar working at 24 GHz, and can obtain multidimensional

information on targets [23]. Soli team of Google designs a 60-GHz radar sensor that can realize fine-grained gesture recognition [24]. Texas Instruments designs a frequency-modulated continuous wave (FMCW) radar sensor at an operating frequency of 76–81 GHz with good performance and can be used to detect vital signs, such as breathing and heart-beat [25]. At present, the Doppler frequency offset of radar signal is widely used in radar gesture recognition. Doppler frequency offset occurs when a motion of a measured object from signal transmission to reception. Kim and Ling [26] used the micro-Doppler radar signal to recognize six human motion states, including running, walking, holding, crawling, boxing, and sitting. Based on the micro-Doppler radar signal, Le et al. [27] realized the classification and recognition of human motion without arm swing. In addition, Seyfioglu et al. [28] analyzed the human motion characteristics in the micro-Doppler spectrum of radar. The above contributions show that it is effective to recognize motion state by using the Doppler frequency offset of radar signal. In addition to the motion recognition based on radar micro-Doppler information, there are other interesting works on motion recognition based on radar 3-D point cloud data or radar echo data. Singh et al. [29] used the radar point cloud data and voxelize them to classify five human movements, including boxing, jumping jacks, jumping, squares, and walking, while Sakamoto et al. [30] used I - Q echo plot to recognize three hand gestures.

As an emerging radar technology, mmWave radar has received widespread attention with regard to its safety issues, human exposure limits, and radiation damage. The mmWave radiation has relatively low energy and weak penetration capability. It is also absorbed very shallowly in human body, generally only in skin surface. Therefore, its negative impact on human body is relatively small in the research conducted so far on the effects of mmWave radiation on human. Currently, the International Electrotechnical Commission (IEC) has developed relevant standards. The IEC standard for mmWave radiation human exposure limit is $100 \mu\text{W}/\text{cm}^2$ [31]. In this article, the IWR1843 produced by Texas Instruments is used as a mmWave sensor with a maximum transmission power of 12 dBm (i.e., 15.8 mW), and the radar radiation azimuth and elevation range are both $(-60^\circ, 60^\circ)$. In our experiment in this article, at a distance of 0.4 m directly in front of the radar, the radiation surface of the radar is a half-sphere of 3 m^2 . Therefore, the radiation power density on this half-sphere is about $0.5241 \mu\text{W}/\text{cm}^2$, which is much lower than the IEC standard.

According to feature types extracted from gesture signals, an appropriate classification algorithm is required. In traditional two-phase classification methods, the empirical features [7], principal component analysis features [7], and sparse features [33] are first extracted and then fed into a classifier, such as support vector machine (SVM), random forest, or decision tree to accomplish dynamic human gesture recognition (HGR). The emerging deep learning algorithm, for example, the CNNs, brings a breakthrough in various applications and potential fields [34], which is regarded as an effective method for dynamic HGR. In [35], a fused CNN

architecture is designed for HGR based on the multistatic radar micro-Doppler signatures. Zhang et al. [36] proposed a CNN approach for dynamic HGR in a driver assistance system. Park et al. [37] used a CNN and long short-term memory (LSTM) to learn the waveforms in a 33-GHz radar-based HGR system. Although great progress has been made in radar sensors and on classification algorithms for dynamic HGR, the main working frequencies of the above radar sensors for HGR are focusing on U-band or lower bands which results in poor recognition accuracy [38]. To improve recognition accuracy, we present an HGR system based on a 77-GHz FMCW radar sensor and TD-CTN.

III. GESTURE RECOGNITION BASED ON DATA CUBE SEQUENCE WITH TD-CNN-TRANSFORMER NETWORK

A. Basic Principle of FMCW Radar

The mmWave radar used in our work emits a continuous chirp signal varying in frequency over time. In radar applications, it is often used in a place as conventional pulsed signal because of its unique advantages. The need for a chirp signal arises from the fact that traditional pulsed signals suffer from Range-Doppler ambiguity. This means that if two targets are located at different ranges but moving at same speed, the radar may not be able to distinguish them. This is because the traditional pulsed radar has a fixed pulse repetition interval (PRI), which limits its ability to measure target velocity accurately. In contrast, a chirp signal with continuous changing frequency allows us to improve range resolution and Doppler sensitivity. By using a chirp signal, the radar can sweep through a range of frequencies and measure reflected signal delay and Doppler shift more precisely. This makes it possible to detect multiple targets at different ranges and speeds, even in a cluttered environment. Another advantage of using chirp signal is its flexibility in terms of facing problems. For example, the chirp rate can be adjusted to suit different applications depending on desired Range and Doppler resolutions. Additionally, a chirp signal can be linear or nonlinear, enabling different types of modulation techniques. The chirp signal emitted by the mmWave radar used in this article is FMCW, which is a sine wave signal whose frequency increases with time. Taking a Single Input Single Output (SISO) radar as an example, a frequency synthesizer generates FMCW signal. When there is no multipath effect and only a single target object is considered, the transmitted and received signals can be expressed as

$$s_t(t) = A_t \exp\left(2\pi\left(f_c t + \frac{k}{2}t^2\right)j\right) \quad (1)$$

$$s_r(t) = A_r \exp\left(2\pi\left(f_c(t - \tau) + \frac{k}{2}(t - \tau)^2\right)j\right) \quad (2)$$

where $t \in [0, T)$, $s_t(t)$ is the transmitted signal, $s_r(t)$ is the received signal, τ represents the time from transmission to reception, k represents the frequency modulation slope of each chirp, f_c is the carrier frequency, A_t represents the amplitude of the transmitted signal, and A_r represents the amplitude of the received signal. The signal reflected by the electromagnetic wave hitting an object is called an echo signal. The echo signal is received by a receiving antenna and then mixed with

the transmitting signal to obtain the intermediate frequency (IF) signal. The frequency of IF signal can be considered as the frequency difference between the received and transmitted signals. IF is calculated as follows [39]:

$$\begin{aligned} s_{\text{IF}}(t) &= f_{\text{LPF}}(s_t(t)s_r^*(t)) \\ &= A_t A_r \exp\left(2\pi\left(f_c \tau - \frac{R}{2}\tau^2 + kt\tau\right)j\right) \end{aligned} \quad (3)$$

where R is the distance from the target to the radar.

1) *Distance Measurement*: According to (3), IF signal frequency is

$$f_{\text{IF}} = \frac{d(f_c \tau - \frac{R}{2}\tau^2 + kt\tau)}{dt} = k\tau. \quad (4)$$

Thus, R can be calculated by

$$R = \frac{c\tau}{2}. \quad (5)$$

Applying (5) to (4), we get

$$f_{\text{IF}} = \frac{2kR}{c} \quad (6)$$

where c is the propagation speed of electromagnetic wave in the air. From (6), we know that the frequency of IF signal is directly proportional to the target distance.

2) *Velocity Measurement*: When to distinguish two objects, if the distance between radar and two objects are same, we need to use the existing difference of their speed to distinguish them. Due to the Doppler effect of an object's speed, IF signals of adjacent chirps will produce phase difference $\Delta\phi$, which is directly proportional to the target speed. $\Delta\phi$ is expressed as

$$\begin{aligned} \Delta\phi &= 2\pi f_{\text{IF}} \Delta\tau = 2\pi f_{\text{IF}} \frac{2\Delta R}{c} \\ &= \frac{4\pi f_{\text{IF}} v T_c}{c} \end{aligned} \quad (7)$$

where v means the average object speed within $\Delta\tau$, T_c is the time between two adjacent chirps, $\Delta\tau$ is the difference between the time when the signals reflected by two objects are received by radar, ΔR is the moving distance of the object in T_c , and $\Delta R \ll R$.

3) *Angle Measurement*: After determining the distance and speed of the measured object, we also need to know the angle of the measured object to distinguish two objects with the same speed and distance from the radar. Simply, in the case where there are one transmitting antenna and two receiving antennas, the transmitting antenna sends a chirp received by the first and the second receiving antennas, respectively. The distance from the first receiving antenna to the object is R , and the distance from the second receiving antenna to the object is $R + \Delta R$. The difference in distance will cause a phase difference between the IF signals received by the two receiving antennas. According to the triangular relationship between R and ΔR , the angle between the distance from the object to the antenna and the vertical direction, denoted as θ , can be expressed as

$$\Delta R = R \sin \theta. \quad (8)$$

The relationship between the angle and phase shift of the IF can be expressed by

$$\Delta\phi' = 2\pi f_{\text{IF}} \frac{2\Delta R}{c} = 2\pi f_{\text{IF}} \frac{2R \sin \theta}{c}. \quad (9)$$

B. Radar Data Cube Sequence

For the received IF signals, the data of each frame is processed to a 3-D tensor. The shape of these tensors is $(S \times C \times A)$, where S is the number of samples in each chirp, C is the number of chirps in each frame, and A is the number of antennas. In order to obtain the distance, velocity, and angle information in the IF signals as mentioned in Section III-A, we perform three FFT operations on the 3-D tensor of the IF signals, i.e., 3-D-FFT. After using 3-D-FFT for each tensor of the IF signals, the result is still a 3-D tensor, which can be considered as a data cube [40]. Since a gesture is a continuous action, the data of each frame is interrelated. Therefore, we concatenate each frame of data as a sequence composed of data cubes, namely, radar data cube sequence. The processing of IF signal to obtain our radar data cube sequence is shown in Fig. 1.

The digitized IF signal is

$$s_{\text{IF}}(n) = \int_{-\infty}^{\infty} s_{\text{IF}}(t) \delta(t - nT_s) dt \quad (10)$$

where $s_{\text{IF}}(n)$ is a chirp in one frame of the IF signal from one receiving antenna. We define the 3-D tensor of each IF signal frame as

$$s_{\text{IF}}(n_s, n_c, n_a) \equiv s_{\text{IF}}(n) \quad (11)$$

where $n_s \in \{1, 2, \dots, S\}$, $n_c \in \{1, 2, \dots, C\}$, $n_a \in \{1, 2, \dots, A\}$, and $s_{\text{IF}}(n_s, n_c, n_a)$ represents the IF signal amplitude value of the n_s th sampling point of the n_c th chirp of the n_a th antenna. In the remainder of this article, the IF signal is represented by $s_{\text{IF}}(n_s, n_c, n_a)$.

1) *Range-FFT for IF Signal*: According to (6), the distance of the measured object is proportional to the frequency of the IF signal

$$\frac{R}{f_{\text{IF}}} = \frac{c}{2k} \Rightarrow R \propto f_{\text{IF}}. \quad (12)$$

Therefore, we perform FFT on N_s sampling points of the n_c th chirp and the n_a th antenna to extract the frequency of the IF signal as

$$S_{\text{IF}}(K_s, n_c, n_a) = \sum_{m=0}^{S-1} s_{\text{IF}}(m, n_c, n_a) W(m) \exp\left(-j \frac{2\pi K_s m}{S}\right) \quad (13)$$

where $W(\cdot)$ represents the Hamming window function. We name this transformation as Range-FFT, i.e., performing S -point FFT transformation on each chirp of each frame of the IF signal.

2) *Doppler-FFT for IF Signal*: According to (7), the speed of the measured object is proportional to the phase offset of the IF signal between the adjacent chirps, i.e.,

$$\frac{v}{\Delta\phi} = \frac{c}{4\pi f_{\text{IF}} T_c} \Rightarrow v \propto \Delta\phi. \quad (14)$$

Therefore, after conducting Range-FFT, we perform FFT between the adjacent chirps of the IF signal to extract the phase offset. This transformation is

$$S_{\text{IF}}(K_s, K_c, n_a) = \sum_{m=0}^{C-1} S_{\text{IF}}(K_s, m, n_a) W(m) \exp\left(-j \frac{2\pi K_c m}{C}\right). \quad (15)$$

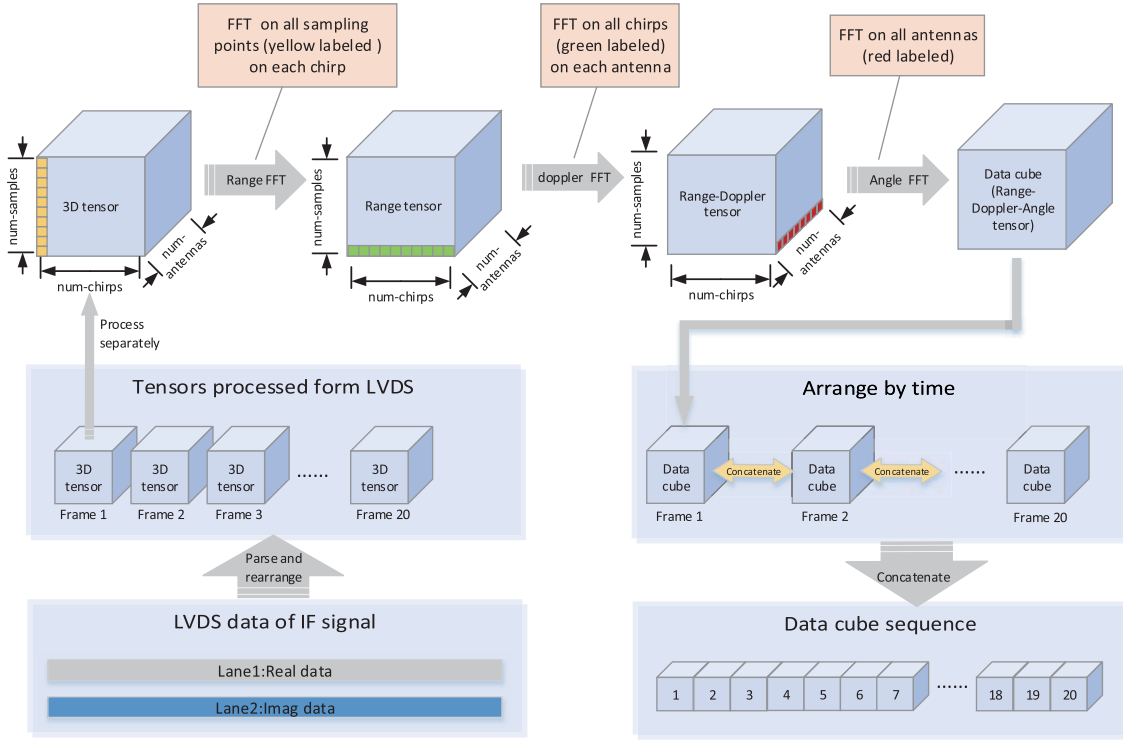


Fig. 1. Process of obtaining radar data cube sequence.

We name this transformation as Doppler-FFT, i.e., performing C -point FFT on each point of the chirp dimension of $S_{IF}(K_s, n_c, n_a)$.

3) *Angle-FFT for IF Signal*: The type of hand gestures can be investigated based on the features of distance, speed, and angle of reflection point cloud from our hands. The aforementioned Range-FFT and Doppler-FFT can only extract distance and velocity information of these reflection points. According to (9), the angle of the measured object is related to the phase offset of the IF signal between the adjacent receiving antennas, i.e.,

$$\frac{\sin \theta}{\Delta \phi'} = \frac{c}{4\pi f_{IF} R} \Rightarrow \sin \theta \propto \Delta \phi'. \quad (16)$$

Therefore, we extract the phase offset between the adjacent antennas to obtain the angle information of the reflection points of hand with FFT. The transformation is calculated by

$$S_{IF}(K_s, K_c, K_a) = \sum_{m=0}^{A-1} S_{IF}(K_s, K_c, m) W(m) \exp\left(-j \frac{2\pi K_a m}{A}\right). \quad (17)$$

We define the above transformation as Angle-FFT, i.e., performing A -point FFT on each point of the antenna dimension of the result after Doppler-FFT transformation. The result of Angle-FFT is saved as a new tensor, called a data cube.

4) *Correlation Between Frames*: Because there are 20 frames of gesture data collected at once, we construct 20 frames of data cube sequence obtained after the processing. Since our gestures are time-dependent sequences, we combine the data cube of 20 frames to obtain a sequence of data cubes arranged in time order. We use X to represent a tensor

of data cube sequence and x_i to represent the i th data cube. For the convenience of representation, we express the frame number of the IF signal with T . Thus, the composition of X is

$$X = [x_1, x_2, \dots, x_i, \dots, x_T], x_i \in \mathbb{R}^{S \times C \times A}. \quad (18)$$

C. CTN Neural Network for Hand Gesture Recognition

We input the data cube sequence into TD-CNN-Transformer as a 4D tensor with shape of $(T \times S \times C \times A)$.

As shown in Fig. 2, the whole CTN can be divided into three segments: 1) convolution layer; 2) position encoding; and 3) transformer layer. The local features of the data cube sequence are extracted through the convolution layer. The position of these local features are first encoded. Then, the transformer extracts the long-distance features of these local features and outputs the classification results.

1) *Convolution Layer in CTN*: The convolution layer is the first segment in the whole CTN. In this segment, we use time distributed wrapper (TD) to encapsulate T parallel convolution layers to process each frame of the data cube sequence, respectively. We extract the local features of each frame of the data cube sequence in the following way. We first take the convolution introduced by [41] in which the value of unit at position (x, y, z) in the j th feature map in the i th layer denoted as α_{ij}^{xyz} is given by

$$\alpha_{ij}^{xyz} = \tanh\left(b_{ij} + \sum_m \sum_{\gamma=0}^{\Gamma_i-1} \sum_{\xi=0}^{\Xi_i-1} \sum_{\nu=0}^{\Upsilon_i-1} \beta_{ijm}^{\gamma\xi\nu} \alpha_{(i-1)m}^{(x+\gamma)(y+\xi)(z+\nu)}\right) \quad (19)$$

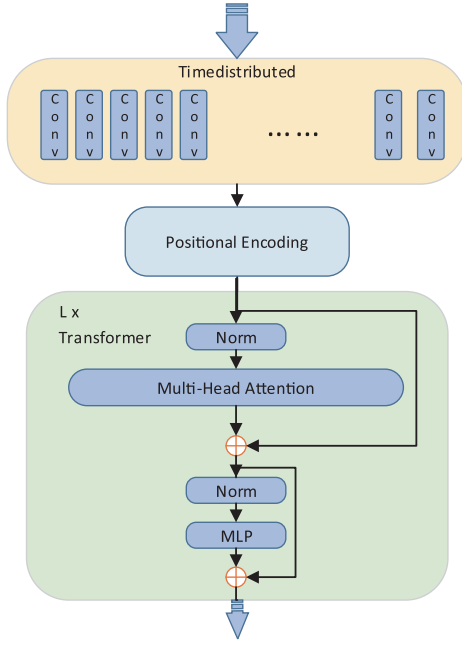


Fig. 2. TD-CNN-transformer structure.

where $\tanh(\cdot)$ is the hyperbolic tangent function, b_{ij} is the bias for this feature map, m indexes over the set of feature maps in the $(i-1)$ th layer connected to the current feature map, $\beta_{ijm}^{\gamma\xi v}$ is the value at the position (γ, ξ, v) of the kernel connected to the k th feature map, and Γ_i and Ξ_i and Υ_i are the height and width and depth of the kernel, respectively. For a simple expression, we use $\text{Conv}(\cdot)$ to express the above convolution. We use y_i to represent the tensor defined by the data cube sequence after the feature extraction from the convolution layer. Thus, each convolution block can be expressed as

$$y_i = \text{Conv}(x_i), \quad i \in \{1, 2, \dots, T\}, \quad (20)$$

$$Y = [y_1, y_2, \dots, y_i, \dots, y_T]. \quad (21)$$

2) *Position Encoding in CTN*: Before the position encoding, we need to change the dimension of the tensors obtained by CNN. We project T tensors y_i into T D -dimensional vectors e_i^o through a flatten layer and a dense layer. We use $\text{Flat}(\cdot)$ and $\text{Dens}(\cdot)$ to represent the flatten and dense functions, respectively. Then, our mapping process is

$$E^o = \text{Dens}([\text{Flat}(y_1), \dots, \text{Flat}(y_i), \dots, \text{Flat}(y_T)]) \quad (22)$$

$$E^o = [e_1^o, e_2^o, \dots, e_i^o, \dots, e_T^o], E^o \in \mathbb{R}^{T \times D}. \quad (23)$$

After obtaining E^o , we calculate the position embedding value for each vector e_i^o in E^o . We use e_i^p to represent the position embedding value corresponding to e_i^o , e_i^p can be expressed as [42]

$$e_i^p = \begin{cases} \sin \Omega_k i, & \text{if } d = 2k, \\ \cos \Omega_k i, & \text{if } d = 2k + 1 \end{cases} \quad (24)$$

$$E^p = [e_1^p, e_2^p, \dots, e_i^p, \dots, e_T^p] \quad (25)$$

where $d \in D$ and Ω_k is

$$\Omega_k = \frac{1}{10000^{2k/d}}. \quad (26)$$

Therefore, the position encoding is

$$E = E^o + E^p, \quad E \in \mathbb{R}^{T \times D} \quad (27)$$

$$E = [e_1, e_2, \dots, e_i, \dots, e_T]. \quad (28)$$

3) *Transformer in CTN*: When tensor E enters L layer transformer encoders, the long-distance features can be extracted. Each layer of the transformer encoder consists of a multihead attention (MA) block and a multilayer perceptron (MLP) block, with the layer normalization applied before each block and the residual connections used after each block [43]. The MA layer is integrated by multiple different self-attention (SA), and each SA has different generation matrices W^q , W^k , and W^v . In this work, we integrate eight SA per MA. The calculation process of each group of SA is as follows. In order to improve the training speed and stability of the neural network, we layer-normalize the tensor after the position encoding as

$$E = LN(E) = \frac{E - \bar{E}}{\sqrt{\text{var}(E) + \epsilon}} \quad (29)$$

where $\bar{E}$ is the mean value of E , $\text{var}(E)$ express the variance of E , ϵ is a small number to prevent the denominator from zero. The encoded tensors $e_i \in E$ are multiplied by W^q , W^k , and W^v , respectively, to obtain the corresponding query vector q_i , key vector k_i and the value vector v_i , i.e.,

$$q_i = e_i \times W^q \quad (30)$$

$$k_i = e_i \times W^k \quad (31)$$

$$v_i = e_i \times W^v \quad (32)$$

$$Q = [q_1, q_2, \dots, q_i, \dots, q_T] \quad (33)$$

$$K = [k_1, k_2, \dots, k_i, \dots, k_T] \quad (34)$$

$$V = [v_1, v_2, \dots, v_i, \dots, v_T]. \quad (35)$$

The attention value can be calculated by using q_i, k_i , and v_i . We calculate a score for the association between the i th encoded vector e_i and the j th encoded vector e_j . The score is denoted as

$$g_{ij} = q_i \times k_j^T, \quad i, j \in \{1, 2, \dots, T\}. \quad (36)$$

The larger g_{ij} is the higher the correlation between e_i and e_j . In order to stabilize the gradient during training, we normalize g_{ij} as

$$g_{ij} = \frac{g_{ij}}{\sqrt{d_k}} \quad (37)$$

where d_k is the dimension of k_j . Then, we use the softmax function $F_{sfm}(\cdot)$ to convert g_{ij} into the probability distribution between $[0, 1]$, and we multiply it by the corresponding v_j . The result of the above calculation is the attention vector of e_i denoted as

$$z_i = \sum_{j=1}^T F_{sfm}\left(\frac{q_i k_j^T}{\sqrt{d_k}}\right) v_j, \quad i \in \{1, 2, \dots, T\}. \quad (38)$$

In this work, 12 different groups of W^q , W^k , and W^v are used to get 12 different z_i . Finally, the obtained z_i are spliced as

$$Z = [z_1, z_2, \dots, z_i, \dots, z_{12}]. \quad (39)$$

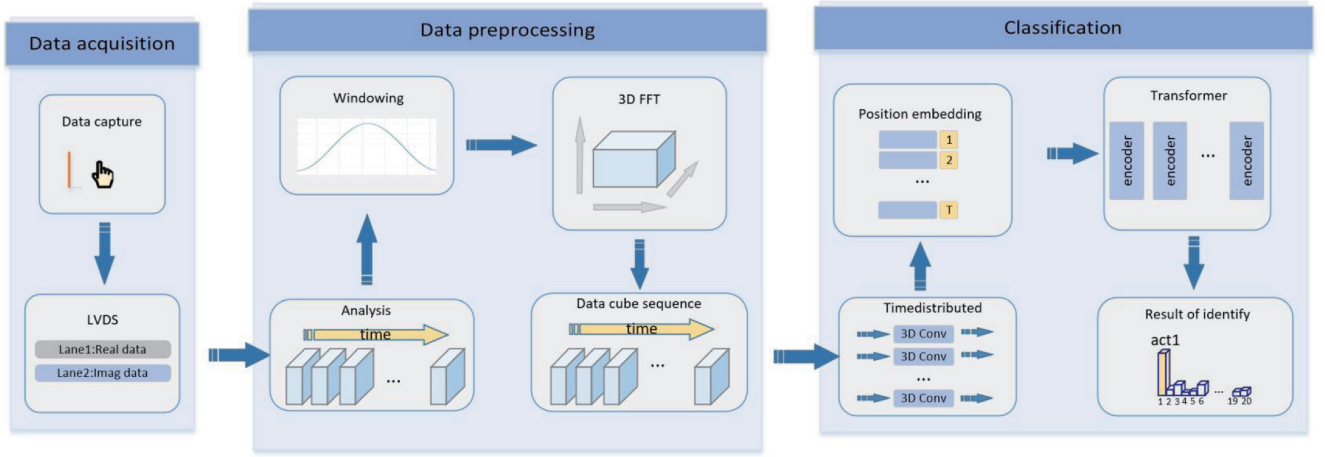


Fig. 3. Structure of hand gesture recognition system.

In order to obtain the same dimension as that of the input encoded vector e_i , we use a weight matrix W to convert the dimension of Z . The calculation result is the output of the whole MA layer

$$Z_{MA} = Z \times W, Z_{MA} \in \mathbb{R}^{T \times D}. \quad (40)$$

After obtaining Z , in order to prevent the degradation in deep neural network training, we define a residual connection

$$Z' = Z_{MA} + E. \quad (41)$$

Z' represents the sequence feature extracted from gesture data. Then, we use MLP for Z' , in order to further extract features. The MLP structure in transformer, also known as feed-forward neural network (FFN), is used in TD-CNN-Transformer to perform a nonlinear transformation on the vector obtained through SA. The MLP used in this article consists of two parts. The first one takes the dense function for linear transformation on the input, and the second layer uses the ReLU activation function for nonlinear transformation on the input. The second part uses the dropout function to prevent the overfitting of the model. The calculation process of the MLP here can be briefly represented as

$$MLP(x) = \text{dropout}(\max(0, xw_1 + b_1)w_2 + b_2) \quad (42)$$

where w_1, b_1, w_2 , and b_2 are the weights and offsets of dense and ReLU functions, respectively. We take the residual connection for the results obtained from MLP as

$$Z_{out} = Z' + MLP(LN(Z')). \quad (43)$$

When Z_{out} is the output of the whole transformer layer, the transformer layer can be cycled. In this work, we establish an algorithm with a loop structure of 8-layer transformers as shown in Algorithm 1.

IV. EXPERIMENTS RESULTS

A. Implementation of DCS-CTN System

As shown in Fig. 3, our proposed gesture recognition system includes three modules: 1) data acquisition; 2) data preprocessing; and 3) gesture classification. Specifically, as shown

Algorithm 1: TD-CNN-Transformer

```

1: Initialization:  $Input \leftarrow [x_1, x_2, \dots, x_i, \dots, x_T], x_i \in \mathbb{R}^{S \times C \times A}, L \leftarrow 0$ 
2:  $Y = [Conv(x_1), \dots, Conv(x_i), \dots, Conv(x_T)]$ 
3:  $E^o = Dens([Flat(y_1), \dots, Flat(y_i), \dots, Flat(y_T)])$ 
4:  $E^p = PositionEmbedding(E^o)$ 
5:  $E = E^o + E^p$ 
6: repeat
7:    $Temp_1 = LN(E)$ 
8:    $Z_{MA} = MultiHeadAttention(Temp_1)$ 
9:    $Z' = E + Z_{MA}$ 
10:   $Temp_2 = LN(Z')$ 
11:   $Temp_3 = MLP(Temp_2)$ 
12:   $Z_{out} = Temp_3 + Z'$ 
13:   $L = L + 1$ 
14:   $E = Z_{out}$ 
15: until  $L \geq 8$ 
16: Output =  $Dens(MLP(Flat(LN(E))))$ 

```

Output: classification results

TABLE I
TIME SPENT IN DIFFERENT STAGES OF DCS-CTN

stage name	time spent per sample
data preprocessing	1.3s-1.4s
classification	0.025s

in Table I, we summarize the processing time per gesture instance at different stages of the DCS-CTN system. As shown in Table I, the data preprocessing stage takes the most time, in which MATLAB code is used. Due to the classification process of the DCS-CTN system implemented on the GPU server, the time spent on this stage is only 0.025 s. In addition to the statistical data in the table, the training time of the TD-CTN model is about 10 s per epoch, and the model trained for 300 epochs can finally converge.

1) *Data Acquisition:* In the experiments, an IWR1843BOOST millimeter-wave radar board and a DCA1000EVM data acquisition board are used to collect the original radar echo data.

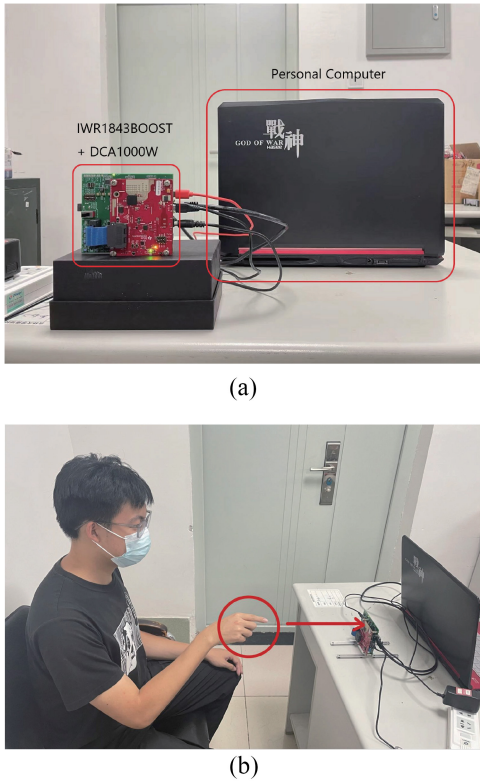


Fig. 4. Experimental equipment. (a) Data acquisition system. (b) Experimental environment.

IWR1843BOOST is a 77-GHz MIMO radar board with three transmitters and four receivers. DCA1000EVM is a capture card interfacing with Texas Instrument's 77 GHz IWR1843BOOST and transmits LVDS from its network port to a host computer under user datagram protocol (UDP) packets. As is shown in Fig. 4, our data acquisition uses IWR1843BOOST and DCA1000EVM at the same time, and we control the acquisition process on a personal computer. The collection process is conducted in a closed room. During the data acquisition period, there are no other moving people or objects in front of the radar. While collecting data, a participant sits in front of the radar with one hand 40 cm in front of the radar antenna and keeps the rest hand still.

2) *Data Preprocessing*: We use a GPU server working on the Windows platform as our host computer with a central processing unit (CPU) of Intel i9-10920x and a graphics card of NVIDIA RTX3080. The received LVDS data is processed on the host computer to restore the data in complex form and rearranged it into a matrix with shape of $(T \times C \times S \times A)$. Then, the matrix is processed with the Hamming windowing, and FFT transformation in range, speed, and angle dimensions. The data cubes generated from the data processing are rearranged into a sequence, i.e., the data cube sequence. Then, we mark the corresponding gesture label for the data cube sequence of each gesture and package it in the form of a compressed file.

3) *Classification*: The compressed file with gesture features is sent to CTN for gesture classification. This CTN consists of three parts: 1) two convolutional layers; 2) a position encoder; and 3) eight transformer layers. The convolutional

TABLE II
CONFIGURATION OF RADAR PARAMETERS

parameter name	value
start frequency	77GHz
FM slope	100MHz/ μ s
ADC sampling points/chirp	256
ADC sampling rate	5209ksps
FM time	50 μ s
FM bandwidth	4MHz
frame period	100ms
chirp counts/frame	16
transmitting antenna	Tx0, Tx1, Tx2
receive antenna	Rx0, Rx1, Rx2, Rx3
transmit power	12dBm
receive antenna gain	36dB
number of frames	20

TABLE III
GESTURE TYPE

Gesture 1:	click with index finger (0.5s-0.8s)
Gesture 2:	double-click with index finger (1.0s-1.6s)
Gesture 3:	clockwise circle with index finger (1.6s-1.8s)
Gesture 4:	counterclockwise circle with index finger (1.6s-1.8s)
Gesture 5:	separate index finger and thumb (0.9s-1.1s)
Gesture 6:	close index finger and thumb (0.9s-1.1s)
Gesture 7:	rub thumb forward with index finger (1.1s-1.3s)
Gesture 8:	rub thumb back with index finger (1.1s-1.3s)
Gesture 9:	draw x with index finger (1.8s-2.0s)
Gesture 10:	draw reverse x with index finger (1.8s-2.0s)
Gesture 11:	draw checkmark with index finger (0.9s-1.1s)
Gesture 12:	shake index finger (1.2s-1.4s)
Gesture 13:	draw index finger upwards (0.9s-1.1s)
Gesture 14:	draw index finger downwards (0.9s-1.1s)
Gesture 15:	draw index finger to the right (0.9s-1.1s)
Gesture 16:	draw index finger to the left (0.9s-1.1s)
Gesture 17:	push the right hand forward (1.8s-2.0s)
Gesture 18:	pull the right hand back (1.8s-2.0s)
Gesture 19:	push both hands forward (1.8s-2.0s)
Gesture 20:	pull both hands back (1.8s-2.0s)

layer is used for the local feature extraction in each time step of the data cube sequence, encapsulated by a TD. The position embedding encodes the position of each time step to retain the timing information of the data. The transformer extracts the long-distance features of the data cube sequence and uses the SA mechanism to extract the correlation and importance of different time steps.

B. Experiment Setup

On the aforementioned system, we conduct comprehensive experiments to evaluate the proposed gesture recognition system. The parameters of the radar used in this article are configured as given in Table II.

In these experiments, each gesture contains 20 frames of data and the duration of each frame is 100 ms. The participant is required to complete a gesture within 2 s. We construct a data set of 20 participants with 20 gestures. Each participant repeats the same gesture five times for a total of 2000 samples. The participants are of ages from 20–30, including nine men and 11 women, using their right hands to make the postures.

The gesture types are shown in Fig. 5 and Table III.

For the comparisons, we also collect our point cloud and Range-Doppler map under the same environment, acquisition device, and participants. In order to ensure the persuasiveness

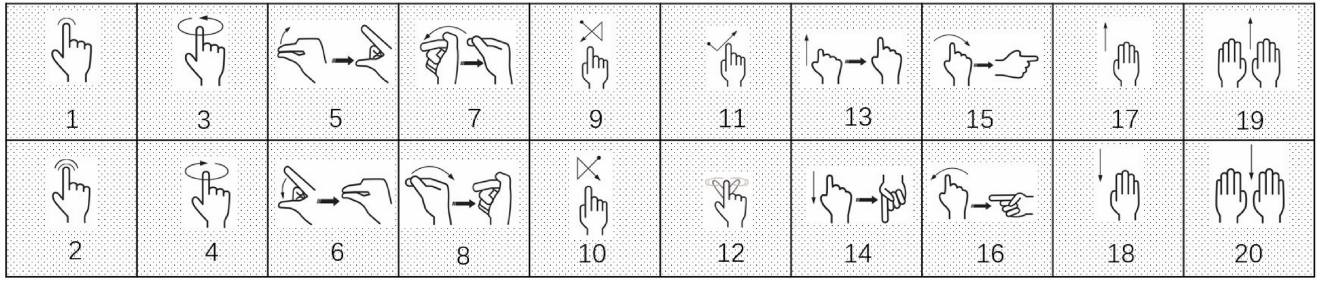


Fig. 5. Twenty gestures.

of the comparisons, we use self-written driver code to realize the simultaneous acquisition of the raw data, point cloud, and Range–Doppler map features. Both the point cloud and the Range–Doppler map are generated by the millimeter-wave radar internal DSP module.

After the gesture acquisition, we analyze and arrange the saved data offline. In our gesture evaluation, one gesture instance corresponds to one label. Before training CTN, we divide all the collected data according to the gesture types. We extract 80% of the data as our training set and 20% of the data as the test set. Our CTN is composed of two layers of CNN and eight layers of transformer with a total of 6 229 108 parameters. The learning rate is set to 0.001, the weight decay to 0.0001, and the batch size is 20, respectively.

C. Evaluation Indicator

When evaluating the performance of CTN in gesture recognition, we mainly use the following parameters, i.e., accuracy, loss, precision, and recall. In this work, the loss of CTN training is the cross-entropy loss defined by

$$\text{loss}(p, q) = - \sum_{i=1}^N p(x_i) \log(q(x_i)) \quad (44)$$

where $p(x_i)$ is the true probability of x_i , and $q(x_i)$ is the fitting probability of x_i . The other used calculation formulas of the evaluation indicators are

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (45)$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (46)$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (47)$$

where true positive (TP) indicates the number of positive samples recognized as positive samples, true negative (TN) indicates the number of negative samples predicted as negative samples, false positive (FP) indicates the number of negative samples recognized as positive samples, and false negative (FN) indicates the number of positive samples recognized as negative samples.

D. Comparison of Different Features Under CNN-LSTM

To investigate the influence of the input feature on the recognition accuracy, we evaluate the obtained point cloud, Range–Doppler heat map, and the proposed data cube sequence

TABLE IV
RECOGNITION ACCURACY OF DIFFERENT FEATURES

feature name	accuracy
point cloud	64.53%
doppler-range	90.73%
data cube sequence	98.51%

TABLE V
RECOGNITION PERFORMANCE OF DIFFERENT FEATURE

	Point cloud		RD heart map		data cube sequence	
	precision	recall	precision	recall	precision	recall
Gesture 1	0.00	0.00	0.47	0.36	1.00	1.00
Gesture 2	0.06	0.05	0.77	0.81	1.00	1.00
Gesture 3	0.11	0.19	0.72	0.62	1.00	1.00
Gesture 4	0.23	0.14	0.59	0.76	1.00	1.00
Gesture 5	0.00	0.00	0.32	0.38	1.00	1.00
Gesture 6	0.20	0.14	0.38	0.24	1.00	1.00
Gesture 7	0.18	0.43	0.40	0.38	1.00	1.00
Gesture 8	0.10	0.05	0.40	0.38	1.00	1.00
Gesture 9	0.09	0.14	0.75	0.86	1.00	1.00
Gesture 10	0.00	0.00	0.71	0.81	1.00	1.00
Gesture 11	0.06	0.14	0.79	0.71	1.00	0.90
Gesture 12	0.15	0.14	0.83	0.71	1.00	1.00
Gesture 13	0.00	0.00	0.48	0.57	1.00	1.00
Gesture 14	0.00	0.00	0.31	0.52	0.95	1.00
Gesture 15	0.12	0.05	0.87	0.62	0.95	1.00
Gesture 16	0.14	0.14	0.56	0.43	1.00	1.00
Gesture 17	0.13	0.10	0.95	0.86	0.95	1.00
Gesture 18	0.09	0.05	0.10	0.81	1.00	0.95
Gesture 19	0.00	0.00	0.91	0.95	1.00	0.95
Gesture 20	0.18	0.33	0.83	0.95	0.95	1.00

based on a commonly used model of CNN-LSTM, which is a spatial–temporal network with a convolution layer to extract local features and an LSTM to extract long-distance features. The evaluation results are shown in Table IV. As shown in Table IV, the average recognition accuracy is about 64.53% under the point cloud, about 90.73% under the Range–Doppler heat map, and around 98.51% under the data cube sequence. From Fig. 6 and Table V, for the recognition performance based on the different input features, we can obtain the following results.

- 1) The recognition performance of the point cloud as the input feature is poor for the subtle gesture recognition, because it is difficult to capture the details of the subtle gestures by the sparse point cloud of mmWave radar.
- 2) When using Range–Doppler map as the input feature, the recognition performance is better than that based on the point cloud, because Range–Doppler map is more intensive data than point clouds. For example, gesture 9 is to draw an X with index finger. Its trajectory is

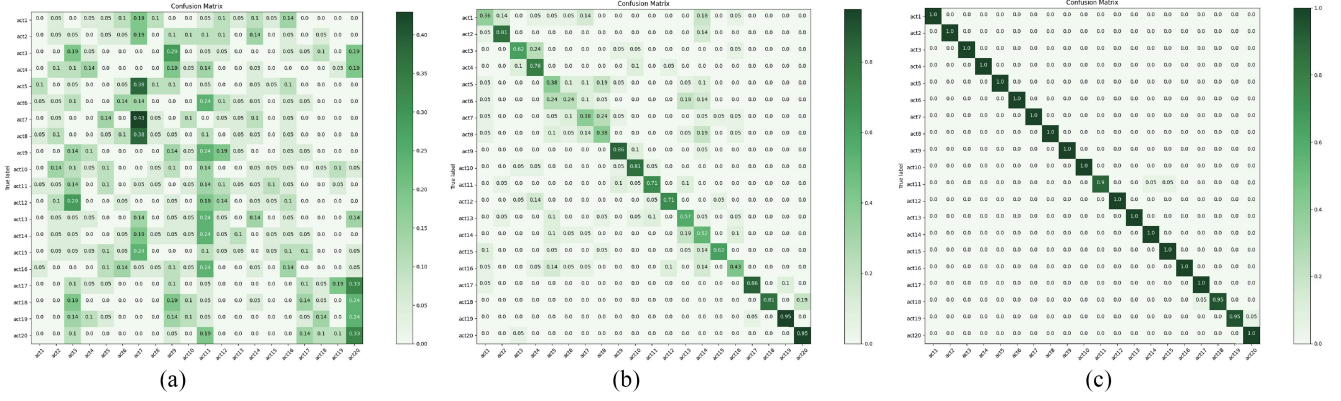


Fig. 6. Confusion matrix of CNN-LSTM with different features. (a) With point cloud. (b) With range-doppler heat map. (c) With data cube sequence.

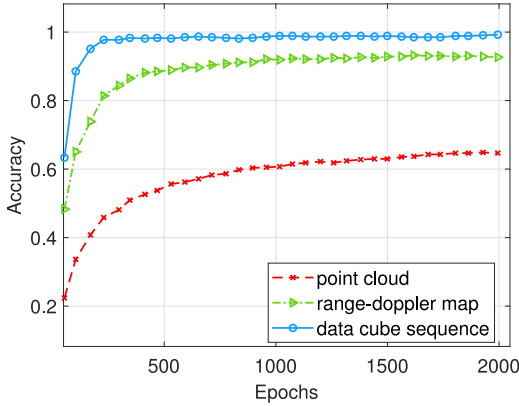


Fig. 7. Accuracy of CNN-LSTM with different features.

complex and 2-D, so it is much better to use a Range-Doppler map as the input feature than a point cloud. The prediction accuracy of the point cloud for gesture 9 is 14%, while that of the Range-Doppler map for gesture 9 is 86%.

- 3) When data cube sequence is adopted as input feature, the recognition accuracy is significantly improved and can achieve about 98.51%, which is much higher than that with the point cloud and Range-Doppler heat map. Such a big improvement is because of the intensive data cube sequence with angle information. Therefore, for some 3-D gestures, higher recognition accuracy can be obtained when the data cube sequence is used as the input feature. For example, gesture 13 is a 3-D gesture in which index finger is swiped upward. When the Range-Doppler map is used, the recognition accuracy of gesture 13 is 57%. When the data cube sequence is used, the recognition accuracy of gesture 13 is almost 100%.

Fig. 7 shows the training accuracy of CNN-LSTM based on these input features. According to Fig. 7, we can draw the following answers.

- 1) Since the point cloud is noisy and sparse, the features of those point clouds cannot accurately represent the subtle gestures. Hence, the convergence speed of CNN-LSTM is low. After 2000 epochs, the model still does not converge.

TABLE VI
ACCURACY OF DIFFERENT NETWORK

algorithm name	prediction accuracy
CNN	97.12%
CNN-LSTM	98.51%
CTN	99.75%

- 2) The convergence speed of CNN-LSTM based on the Range-Doppler map is much higher than that of the point cloud. When the network training is about 100 epochs, the recognition accuracy reaches around 80%, and then it is gradually stable.
- 3) When using the data cube sequence as the input feature, the model of CNN-LSTM reaches 98% after 100 epochs, which is much faster than that of using the point cloud and the Range-Doppler map, because the data cube sequence can better represent these subtle gestures more accurately.

E. Comparison of Different Network

We compare the performance of the gesture recognition using different models, i.e., the proposed CTN, CNN-LSTM, and CNN, respectively. The structure of CNN is of three-stage, each of which is composed of two convolution layers and one max-pool layer. The gesture recognition accuracy is statistically summarized in Table VI.

In this table, we can find that using the data cube sequence as the input feature, the recognition accuracies of CNN, CNN-LSTM, and CTN are about 97.12%, 98.51%, and 99.75%, respectively. In order to investigate different gestures under different neural networks, we provide the confusion matrix shown in Fig. 8 and summarize the precision and recall under the several data sets as shown in Table VII.

According to Fig. 8 and Table VII, we can give the following results.

- 1) In the recognition results based on CNN, there occur errors at least 5% in the prediction of almost half of the gestures, such as gesture 1, gesture 7, gesture 12, gesture 13, gesture 14, gesture 18, and gesture 20. Among them, the prediction effect of gesture 12 is the

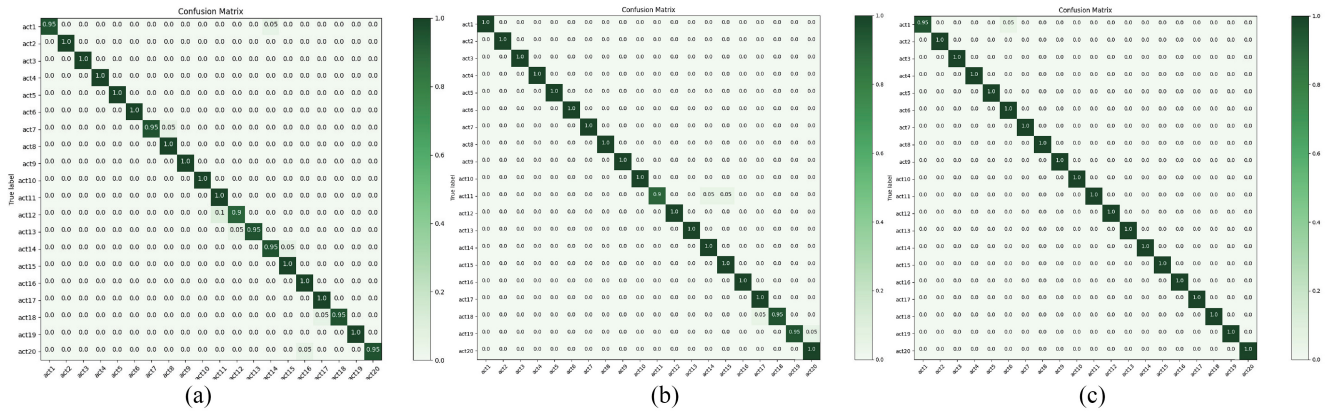


Fig. 8. Confusion matrix of different networks with data cube sequence. (a) CNN. (b) CNN-LSTM network. (c) CTN.

TABLE VII
RECOGNITION PERFORMANCE OF DIFFERENT NETWORKS

	CNN		CNN-LSTM		CTN	
	precision	recall	precision	recall	precision	recall
Gesture 1	1.00	0.95	1.00	1.00	1.00	0.95
Gesture 2	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 3	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 4	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 5	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 6	1.00	1.00	1.00	1.00	0.95	1.00
Gesture 7	1.00	0.95	1.00	1.00	1.00	1.00
Gesture 8	0.95	1.00	1.00	1.00	1.00	1.00
Gesture 9	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 10	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 11	0.91	1.00	1.00	0.90	1.00	1.00
Gesture 12	0.95	0.90	1.00	1.00	1.00	1.00
Gesture 13	1.00	0.95	1.00	1.00	1.00	1.00
Gesture 14	0.95	0.95	0.95	1.00	1.00	1.00
Gesture 15	0.95	1.00	0.95	1.00	1.00	1.00
Gesture 16	0.95	1.00	1.00	1.00	1.00	1.00
Gesture 17	0.95	1.00	0.95	1.00	1.00	1.00
Gesture 18	1.00	0.95	1.00	0.95	1.00	1.00
Gesture 19	1.00	1.00	1.00	0.95	1.00	1.00
Gesture 20	1.00	0.95	0.95	1.00	1.00	1.00

worst at about 90%. The reason is that this CNN does not take advantage of the temporal continuity of the gestures.

- 2) When we take the CNN-LSTM network, there are fewer prediction errors, 5% errors occur in gesture 11, gesture 18, and gesture 19. It shows that on the basis of CNN, the addition of LSTM further improves the accuracy of gesture recognition, since it brings the ability to extract time sequence information to the model.
- 3) When using the position encoding and the transformer, the recognition accuracy also increases. There are only 5% errors occurring in gesture 1 under CTN. Compared with CNN-LSTM, CTN can help us to improve the accuracy after using the transformer instead of LSTM, because the model can distinguish the data importance of the gestures at different time, and each moment is not of same importance for the gestures. For example, the moment of a finger click in gesture 1 has a greater impact on the correct recognition of the gesture than other times.

Moreover, we present the recognition accuracy results under various gesture recognition networks shown in

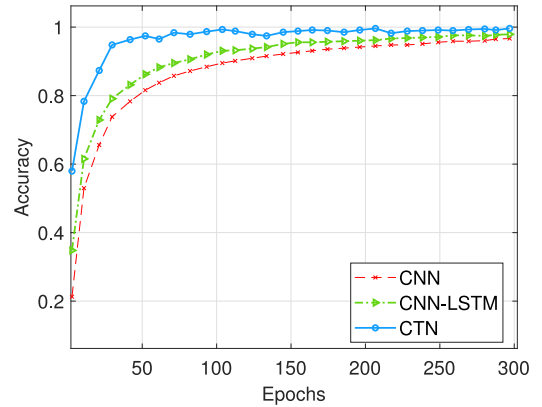


Fig. 9. Accuracy of different networks with data cube sequence.

Fig. 9, with batch size of 20 when using three given networks. Following Fig. 9, we can find the following answers.

- 1) From Fig. 9, it can be seen that when using CNN and the data cube sequence for gesture recognition, the recognition accuracy already reaches a high level. This is due to the rich gesture information contained in the data cube sequence. However, the convergence speed is lower, because CNN cannot use the timing information of the data cube sequence.
- 2) Compared with CNN, CNN-LSTM has little higher convergence speed and accuracy, since CNN-LSTM can recognize timing information of the gestures. After training in 50 epochs, the convergence rate of CNN and CNN-LSTM are all slow and gradually stabilize.
- 3) When CTN is used, the convergence speed increases with light fluctuation, and the accuracy is also slightly better. CTN can reach the peak accuracy within 20 epochs, and the accuracy is stable at 99% after about 100 epochs. Therefore, we can say that this network has a faster convergence speed and higher recognition accuracy compared with CNN and CNN-LSTM. The reason for these improvements is that the introduced transformer can distinguish the importance of the gestures at different times.

TABLE VIII
ACCURACY OF CTN WITH DIFFERENT TRANSFORMER LAYERS

algorithm name	prediction accuracy
CTN(4×transformer)	99.75%
CTN(8×transformer)	99.75%
CTN(12×transformer)	99.75%

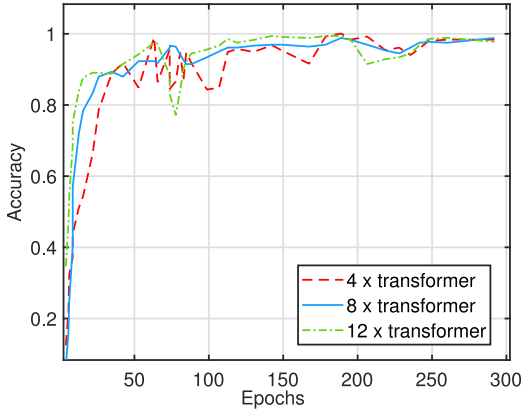


Fig. 10. Accuracy of CTN with different layer transformer.

F. Comparison of CTN With Different Transformer Layers

For CTN with different layers of transformers, we summarize the accuracy of the gesture classification shown in Table VIII.

The average accuracy of this gesture classification is the same as for CTN with 4, 8, and 12 layers of transformers, respectively.

As shown in Fig. 10 and Table IX, we also present the classification accuracy of CTN for different categories of the gestures in these cases. The results show that the accuracy rate is the same as those in the three cases. The model converged along epochs fluctuates in different ranges with the number of transformer layers according to Fig. 11. CTN with 8-layer transformers has the most stable accuracy. When the layers are deeper, the ability of CTN for feature extraction is stronger. However, it will be difficult to train the network with too many layers. Therefore, we need to properly set the number of transformer layers in CTN.

G. Comparison of Different Scenarios

We further conduct additional experiments in different scenarios to evaluate the influence of comparison factors on the system. The results are shown in Table X.

- 1) We compare the accuracy of the gesture recognition under different interference by setting up a scene with a second participant in the radar scanning area and a scene where another hand of the participant randomly moves. The recognition results show that the accuracy is affected to some extent when there are other people or moving objects in the scenes, especially when the interference object is not stationary, which produces the greatest impact on gesture recognition.
- 2) We compare the influence of fast and slow motion of gestures on the accuracy of gesture recognition by setting up two scenes with 20 kinds of gestures performed

TABLE IX
RECOGNITION PERFORMANCE OF CTN WITH DIFFERENT LAYER TRANSFORMER

	4×transformer		8×transformer		12×transformer	
	precision	recall	precision	recall	precision	recall
Gesture 1	1.00	0.95	1.00	1.00	1.00	0.95
Gesture 2	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 3	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 4	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 5	0.95	1.00	1.00	1.00	1.00	1.00
Gesture 6	1.00	1.00	1.00	1.00	0.95	1.00
Gesture 7	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 8	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 9	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 10	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 11	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 12	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 13	1.00	1.00	0.95	1.00	1.00	1.00
Gesture 14	1.00	1.00	1.00	0.95	1.00	1.00
Gesture 15	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 16	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 17	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 18	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 19	1.00	1.00	1.00	1.00	1.00	1.00
Gesture 20	1.00	1.00	1.00	1.00	1.00	1.00

TABLE X
ACCURACY OF CTN IN DIFFERENT SCENARIOS

scenario description	prediction accuracy
there are others in front of the radar	87.50%
the second hand is moving	83.75%
hand movements are faster	98.75%
hand movements are slower	99.50%
male gestures	99.80%
female gestures	99.65%
the pitch angle from hand to radar is 30°	99.75%
the azimuth angle from hand to radar is 30°	99.75%

at fast speed (within 1 s) and slow speed (within 3 s), respectively. The experimental results demonstrate that our gesture recognition system can reach high accuracy in both fast and slow scenes.

- 3) Then, we compare the influence of hand size on the accuracy of the gesture classification through dividing the data into two groups according to participant gender (men generally have larger hands than women). The results of gesture classification show that the size of hand has little effect on the classification accuracy.
- 4) Finally, we compare the influence of relative position between the participant and radar on the classification accuracy. Since the horizontal and vertical radiation range of the radar used in this article are both $(-60^\circ, 60^\circ)$, we set up the scenes with a participant's hand on the right side of the radar at 30° and above the radar at 30° . From the experimental results, we can draw the conclusion that within the radiation range of the radar, changing relative angle of entire hand to the radar has little effect on the gesture recognition accuracy of the DCS-CTN system.

V. CONCLUSION

In this work, we provide a privacy-protected and contactless subtle gesture recognition system based on a 77–81-GHz FMCW radar, namely, DCS-CTN system. In this system, we

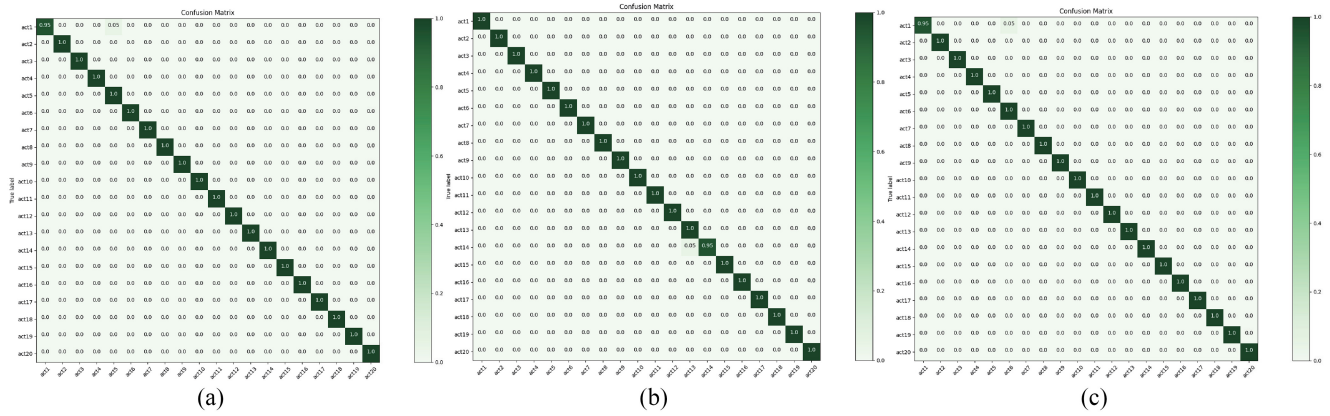


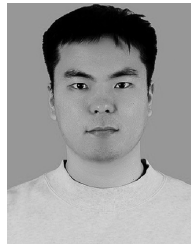
Fig. 11. Confusion matrix of CTN with different layer transformer. (a) CNN-4 $\times$ transformer. (b) CNN-8 $\times$ transformer. (c) CNN-12 $\times$ transformer.

use the radar data cube sequence with high data density as the input feature. In order to improve the accuracy and stability of gesture recognition of the system, we use TD-CNN neural network to extract local features of each data cube. Then, we use a position encoder to encode the extracted features for keeping time information, and we take a transformer neural network to extract global features of the whole sequence and output the results. Through experimental comparison, we find that when CNN-LSTM neural network is used for the gesture recognition the accuracy using data cube sequence as our input feature is about 7.78% higher than that of using Range-Doppler heat map as the input feature, and 33.98% higher than that of using point cloud as the input feature, by reaching 98.51%. We also conclude that the system with the data cube sequence as input and TD-CNN-transformer for the recognition can make the gesture recognition accuracy around 2.63% higher than that of using CNN network, and 1.24% higher than that of using CNN-LSTM network, which can attain almost 99.75%. In addition, compared with gesture recognition based on vision, this proposed DCS-CTN system also has good privacy protection and is not affected by lighting conditions. Therefore, DCS-CTN can be used for contactless subtle gesture recognition in wide applications.

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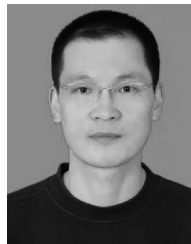
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